Introduction

BCSE304L

Dr. Adrija Bhattacharya
VIT, Vellore

Methods of proof

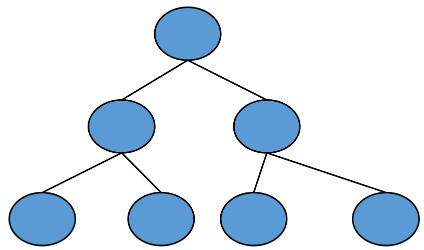
- A theorem is a statement that can be shown to be true.
- A proof is to demonstrate that a theorem is true with a sequence of statements that form an argument.
- An axiom or postulate is the underlying assumption about mathematical structures, the hypothesis of the theorem to be proved, and previously proved theorems.
- The rules of inference are the means used to draw conclusion from other assertions which tie together the steps of a proof.
- A lemma is a simple theorem used in the proof of other theorems.
- A corollary is a proposition that can be established directly from a theorem that has been proved.
- A conjecture is a statement whose truth is unknown. When its proof is found, it becomes a theorem.

On Theorems, Lemmas and Corollaries

We typically refer to:

- A major result as a "**theorem**"
- An intermediate result that we show to prove a larger result as a "**lemma**"
- A result that follows from an already proven result as a "**corollary**"

An example:
Theorem: *The height of an n-node binary tree is at least floor(lg n)*
Lemma: *Level i of a perfect binary tree has 2^i nodes.*
Corollary: *A perfect binary tree of height h has 2^{h+1}-1 nodes.*



Deductive Proofs

From the given statement(s) to a conclusion statement (what we want to prove)

- Logical progression by direct implications

Example for parsing a statement:

- "If $y \geq 4$,   then $2^y \geq y^2$."

given   *conclusion*

(there are other ways of writing this).

Example: Deductive proof

Let <u>Claim 1:</u> If $y \geq 4$, then $2^y \geq y^2$.

Let x be any number which is obtained by adding the squares of 4 positive integers.

<u>Claim 2:</u>

Given x and assuming that Claim 1 is true, prove that $2^x \geq x^2$

- <u>Proof:</u>
  1) Given: $x = a^2 + b^2 + c^2 + d^2$
  2) Given: $a \geq 1$, $b \geq 1$, $c \geq 1$, $d \geq 1$
  3) ➔ $a^2 \geq 1$, $b^2 \geq 1$, $c^2 \geq 1$, $d^2 \geq 1$ (by 2)
  4) ➔ $x \geq 4$ (by 1 & 3)
  5) ➔ $2^x \geq x^2$ (by 4 and Claim 1)

"implies" or *"follows"*

Quantifiers

"For all" or *"For every"*

- Universal proofs
- Notation= $\forall$

"There exists"

- Used in existential proofs
- Notation= $\exists$

Implication is denoted by =>

- E.g., "IF A THEN B" can also be written as "A=>B"

Different ways of saying the same thing

- "*If* H *then* C":
  i. H *implies* C
  ii. *H => C*
  iii. C *if* H
  iv. H *only if* C
  v. *Whenever* H *holds*, C *follows*

“If-and-Only-If” statements

- “A if and only if B” (A <==> B)
  - *(if part)* if B then A (<=)
  - *(only if part)* A only if B (=>)
    (same as “if A then B”)
- “If and only if” is abbreviated as “iff”
  - i.e., “A iff B”
- <u>Example:</u>
  - <u>Theorem:</u> *Let x be a real number. Then floor of x = ceiling of x <u>if and only if</u> x is an integer.*
- Proofs for iff have two parts
  - One for the “if part” & another for the “only if part”

Rules of Inference

The rules of inference provide the justification of the steps used to show that a conclusion follows logically from a set of hypotheses.

- Basis of the rule of inference: the law of detachment

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

hypotheses (p, $p \rightarrow q$)

conclusion ($\therefore q$)

It is the same as to say that $(p \wedge (p \rightarrow q)) \rightarrow q$ is a tautology.

Example 1

It is snowing today.

If it snows today, then we will go skiing.

We will go skiing.

Rules of Inference

Rules of Inference	Tautology	Name
$\dfrac{p}{\therefore p \vee q}$	$p \rightarrow (p \vee q)$	Addition
$\dfrac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Simplification
$\dfrac{\begin{array}{l} p \\ q \end{array}}{\therefore p \wedge q}$	$(p \wedge q) \rightarrow (p \wedge q)$	Conjunction
$\dfrac{\begin{array}{l} p \\ p \rightarrow q \end{array}}{\therefore q}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens (detachment law)
$\dfrac{\begin{array}{l} \neg q \\ p \rightarrow q \end{array}}{\therefore \neg p}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\dfrac{\begin{array}{l} p \rightarrow q \\ q \rightarrow r \end{array}}{\therefore p \rightarrow r}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\dfrac{\begin{array}{l} p \vee q \\ \neg p \end{array}}{\therefore q}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism

Example 2

If n is divisible by 3, then n^2 is divisible by 9.
n is divisible by 3.

$\therefore n^2$ is divisible by 9.

Example 3

State which rule of inference is the basis of the following argument : It is below freezing now. Therefore, it is either below freezing or raining now.

Solution

$$\frac{p}{\therefore p \vee q}$$

Example 4

State which rule of inference is the basis of the following argument : It is below freezing and raining now. Therefore, it is below freezing now.

Solution

$$\frac{p \wedge q}{\therefore p}$$

Example 5

State which rule of inference is the basis of the following argument: If it rains, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow. Therefore, if it rains today, we will have a barbecue tomorrow.

Solution

$$\begin{array}{l} p \to q \\ q \to r \\ \hline \therefore p \to r \end{array}$$

Example 6

p: It is sunny this afternoon.

Show the following hypotheses:

(1) It is not sunny this afternoon and it is colder than yesterday. *q*

(2) We will go swimming only if it is sunny. *r*

(3) If we do not go swimming, then we will take a canoe trip. *s*

(4) If we take a canoe trip, then we will be home by sunset. *t*

lead to the conclusion "We will be home by sunset".

Step	Reason
1. $\neg p \wedge q$	Hypothesis
2. $\neg p$	Simplification (1)
3. $r \to p$	Hypothesis
4. $\neg r$	Modus tollens (2,3)
5. $\neg r \to s$	Hypothesis
6. s	Modus ponens (4,5)
7. $s \to t$	Hypotheis
8. t	Modus ponens (6,7)

Example 7

Show that the hypotheses

(1) If you send me an e-mail message, p

then I will finish writing the program, q

(2) If you do not send me an e-mail massage, then I will go to sleep early, r

(3) If I go to sleep early, then I will wake up feeling refreshed s

lead to the conclusion "If I do not finish writing program, then I will wake up feeling refreshed".

Step	Reason
1. $p \to q$	Hypothesis
2. $\neg q \to \neg p$	Contraposition (1)
3. $\neg p \to r$	Hypothsis
4. $\neg q \to r$	Hypothetical syllogism (2,3)
5. $r \to s$	Hypothsis
6. $\neg q \to s$	Hypothetical syllogism(4,5)

An argument is called *valid* if whenever hypotheses are true, the counclution is also true.
Consequently, q logically follows from the hypotheses $p_1, p_2, \ldots, p_n$ is the same as showing that the implication $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \to q$ is true.

Proving techniques

- By induction
  - (3 steps) Basis, inductive hypothesis, inductive step
- By contradiction
  - Start with the statement contradictory to the given statement
  - E.g., To prove (A => B), we start with:
    - (A and ~B)
    - … and then show that could never happen

    What if you want to prove that "(A and B => C or D)"?
- By contrapositive statement
  - If *A* then *B* ≡ If ~*B* then ~*A*

Proving techniques…

- By counter-example
  - Show an example that disproves the claim

- Note: There is no such thing called a "proof by example"!
  - So when asked to prove a claim, an example that satisfied that claim is not a proof

Induction

•If we have a propositional function P(n), and we want to prove that P(n) is true for any natural number n, we do the following:

- **Show that P(0) is true.**
  (basis step)
- **Show that if P(n) then P(n + 1) for any n$\in$N.**
  (inductive step)
- **Then P(n) must be true for any n$\in$N.**
  (conclusion)

Induction

- **Example:** Show that $n < 2^n$ for all positive integers n.
- Let P(n) be the proposition "$n < 2^n$."
- 1. Show that P(1) is true.
(basis step)
- P(1) is true, because $1 < 2^1 = 2$.

Induction

- 2. Show that if P(n) is true, then P(n + 1) is true. (inductive step)

- Assume that $n < 2^n$ is true.
- We need to show that P(n + 1) is true, i.e.
- $n + 1 < 2^{n+1}$

- We start from $n < 2^n$:
- $n + 1 < 2^n + 1 \leq 2^n + 2^n = 2^{n+1}$
- Therefore, if $n < 2^n$ then $n + 1 < 2^{n+1}$

Induction

3. Then P(n) must be true for any positive integer. (conclusion)

- $n < 2^n$ is true for any positive integer.
- End of proof.

Induction

- **Another Example ("Gauss"):** 1 + 2 + ... + n = n (n + 1)/2

1. Show that P(0) is true.
   (basis step)

- For n = 0 we get 0 = 0. True.

Induction

2. Show that if P(n) then P(n + 1) for any $n \in N$. (inductive step)

- $1 + 2 + \ldots + n = n (n + 1)/2$
- $1 + 2 + \ldots + n + (n + 1) = n (n + 1)/2 + (n + 1)$
- $= (2n + 2 + n (n + 1))/2$
- $= (2n + 2 + n^2 + n)/2$
- $= (2 + 3n + n^2)/2$
- $= (n + 1) (n + 2)/2$
- $= (n + 1) ((n + 1) + 1)/2$

Induction

3. Then P(n) must be true for any $n \in N$. (conclusion)

- $1 + 2 + \ldots + n = n(n + 1)/2$ is true for all $n \in N$.
- End of proof.

Induction

- There is another proof technique that is very similar to the principle of mathematical induction.

- It is called **the second principle of mathematical induction**.

- It can be used to prove that a propositional function P(n) is true for any natural number n.

Induction

- The second principle of mathematical induction:

- Show that P(0) is true.
  (basis step)
- Show that if P(0) and P(1) and … and P(n), then P(n + 1) for any $n \in N$.
  (inductive step)
- Then P(n) must be true for any $n \in N$.
  (conclusion)

Induction

•Example: Show that every integer greater than 1 can be written as the product of primes.

- Show that P(2) is true.
  (basis step)

•2 is the product of one prime: itself.

Induction

- Show that if P(2) and P(3) and … and P(n), then P(n + 1) for any $n \in N$. (inductive step)

- Two possible cases:
- If (n + 1) is **prime**, then obviously P(n + 1) is true.
- If (n + 1) is **composite**, it can be written as the product of two integers a and b such that
  $2 \leq a \leq b < n + 1$.
- By the **induction hypothesis**, both a and b can be written as the product of primes.
- Therefore, $n + 1 = a \cdot b$ can be written as the product of primes.

Induction

- Then P(n) must be true for any $n \in N$. (conclusion)

•End of proof.

•We have shown that **every integer greater than 1** can be written as the product of primes.

Why does Mathematical induction work?

$$
P(1) \qquad \begin{array}{l} P(1) \\ \underline{P(1) \to P(2)} \\ \therefore P(2) \end{array} \qquad \begin{array}{l} P(2) \\ \underline{P(2) \to P(3)} \\ \therefore P(3) \end{array} \qquad \begin{array}{l} P(3) \\ \underline{P(3) \to P(4)} \\ \therefore P(4) \end{array} \quad \cdots\cdots\cdots \quad \begin{array}{l} P(k-1) \\ \underline{P(k-1) \to P(k)} \\ \therefore P(k) \end{array}
$$

$$
\begin{array}{l} p \\ \underline{p \to q} \\ \therefore q \end{array}
$$

Law of detachment

Example 1

P(n)

Use mathematical induction to prove that the sum of the first n odd positive integers is n^2.

Basis step : We prove P(1) is true. $1 = 1^2$. Therefore, P(1) is true.

Inductive step : Assuming *P(k)* is true, we prove $P(k+1)$ is true.

$$1 + 3 + 5 + \ldots + (2k - 1) + (2k + 1)$$

$$= k^2 + (2k + 1) = (k + 1)^2$$

Therefore, for any k if *P(k)* is true then $P(k+1)$ is true.

Example 2

P(n)

Use Mathematical induction to prove the inequality

$n < 2^n$ for all positive integers n.

Basis step : $P(1)$ is true, since $1 < 2^1$.

Inductive step : Assume $P(k)$ is true, that is, $k < 2^k$. We need to show that $P(k+1)$ is true.

From $k+1 < 2^k + 1 \leq 2^k + 2^k = 2^{k+1}$, we get that $P(k+1)$ is true.

Mathematical induction is used to prove the form $\forall n P(n)$ is true.

Where the universe of discourse is the set of contiguous integers: *m,m+1,m+2,….*

1. Basis step. The proposition P(*m*) is shown to be true.
2. Inductive step. The implication

$$P(k) \rightarrow P(k+1) \text{ is shown to be true for every positive integer } k \geq m.$$

Example 3

P(n)

Use Mathematical induction to prove that

$$1+2+2^2+2^3+\cdots+2^k = 2^{k+1}-1 \text{ for all nonnegative integers } k.$$

Basis step: $P(0)$ is true since $2^0 = 1 = 2^1 - 1$.

Inductive step: Assuming P(k) is true, we prove P(k+1) is true:

$$1+2+2^2+2^3+\cdots+2^k+2^{k+1}$$
$$=(1+2+2^2+2^3+\cdots+2^k)+2^{k+1}$$
$$=(2^{k+1}-1)+2^{k+1} = 2\cdot 2^{k+1}-1 = 2^{k+2}-1.$$

Example 4   Sums of Geometric Progressions

Use Mathematical induction to prove the following formula:

$$\sum_{j=0}^{n} ar^j = a + ar + ar^2 + \cdots ar^n = \frac{ar^{n+1} - a}{r - 1}, \text{ when } r \neq 1.$$

P(n)

$$\textit{Basis step}: P(0) \text{ is true, since } a = \frac{ar - a}{r - 1}$$

$$\textit{Inductive step}: \textit{Assume } P(k) \text{ is true. We prove } P(k+1) \text{ is true}:$$

$$a + ar + ar^2 + \cdots ar^k + ar^{k+1}$$

$$= (a + ar + ar^2 + \cdots ar^k) + ar^{k+1} = \frac{ar^{k+1} - a}{r - 1} + ar^{k+1} = \frac{ar^{k+2} - a}{r - 1}$$

Example 5   Inequality for Harmonic Numbers.

$$\text{(Hormonic numbers}: H_i = 1 + \frac{1}{2} + \frac{2}{3} + \cdots + \frac{1}{i},\ \text{for } i = 1,2,3,....)$$

Use Mathematical induction to prove that $H_{2^n} \geq 1 + n/2.$ P(n)

Basis step: $P(0)$ is true, since $H_{2^0} = H_1 = 1 \geq 1 + 0/2.$

Inductive step: *Assume* $P(k)$ is true. We prove $P(k+1)$ is true:

$$H_{2^{k+1}} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^k} + \frac{1}{2^k+1} + \cdots + \frac{1}{2^{k+1}}$$

$$= H_{2^k} + \frac{1}{2^k+1} + \cdots + \frac{1}{2^{k+1}}$$

$$\geq (1 + \frac{k}{2}) + \frac{1}{2^k+1} + \cdots + \frac{1}{2^{k+1}} \text{ (by the inductive hypothesis)}$$

$$\geq (1 + \frac{k}{2}) + 2^k \cdot \frac{1}{2^{k+1}} \text{ (since there are terms each not less than } 1/2^{k+1})$$

$$\geq (1 + \frac{k}{2}) + \frac{1}{2} = 1 + \frac{k+1}{2}.$$

Example 6 The numbers of Subsets of a Finite Set

Use Mathematical induction to prove that

If S is a finite set with n elements, then S has 2^n subsets.

P(n)

Basis step : $P(0)$ is true, since the empty set has exactly

$2^0 = 1$ subsets, namely, itself.

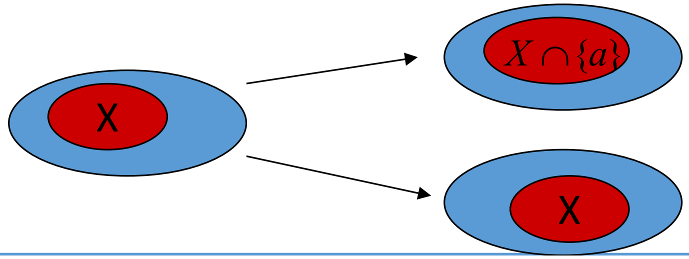
Inductive step : *Assume* $P(k)$ is true. We prove that $P(k+1)$ is true :

Let a be one element of S and $S' = S\text{-}\{a\}$.

The subsets of S can be obtained in the following way :

For each subset X *of* S', X and $X \cup \{a\}$ are the subsets of S.

Since there are 2^k subsets of S, therefore S has $2 \cdot 2^k = 2^{k+1}$ subsets.



Form of Proof by Contraposition

- Theorem: $p \to q$.
- This is logically equivalent to $\sim q \to \sim p$.
- Outline of the proof of the theorem:
  - Assume $\sim q$.
  - Prove $\sim p$.
  - Conclude that $p \to q$.
- This is a *direct* proof of the contrapositive.

Benefit of Proof by Contraposition

- If p and q are negative statements, then $\sim p$ and $\sim q$ are positive statements.
- We may be able to give a *direct* proof that $\sim q \rightarrow \sim p$ more easily that we could give a direct proof that $p \rightarrow q$.

Example: Proof by Contraposition

- Theorem: The sum of a rational and an irrational is irrational.
- Restate the theorem: Let r be a rational number and let α be a number. If α is irrational, then $r + \alpha$ is irrational.
- Restate again: Let r be a rational number and let α be a number. If $r + \alpha$ is rational, then α is rational.

The Proof

- Proof:
  - Let r be rational and α be a number.
  - Suppose that $r + \alpha$ is rational.
  - Let $s = r + \alpha$. Then $\alpha = s - r$, which is rational.
  - Therefore, if $r + \alpha$ is rational, then α is rational.
  - It follows that if α is irrational, then $r + \alpha$ is irrational.

Example

To prove

$$\boxed{\text{If } 3n+2 \text{ is odd, then } n \text{ is also odd.}}$$

we can prove the equivalent statement in **contrapositive form**:

or $$\boxed{\text{If } n \text{ is not odd, then } 3n+2 \text{ is not odd.}}$$

$$\boxed{\text{If } n \text{ is even, then } 3n+2 \text{ is even.}}$$

Rationale

Why?

$$\boxed{\text{If } 3n+2 \text{ is odd, then } n \text{ is also odd.}}$$

$$\boxed{\text{If } n \text{ is not odd, then } 3n+2 \text{ is not odd.}}$$

Recall: Negation

Statement: *n* is odd

Negation of the statement: *n* is **not** odd

Or: *n* is even

Recall: Negation

Notations

$$\boxed{\begin{array}{l} \quad P\text{: } n \text{ is odd} \\ \sim P\text{: } n \text{ is not odd} \end{array}}$$

Contrapositive

The **contrapositive form** of

$$\boxed{\text{If } P \text{ then } Q}$$

is

$$\boxed{\text{If } \sim Q \text{ then } \sim P}$$

Example 4

$\boxed{\text{If } 3n+2 \text{ is odd, then } n \text{ is also odd.}}$

Analysis	Proof: We prove the contrapositive:
	$\boxed{\text{If } n \text{ is even, then } 3n+2 \text{ is even.}}$

Contrapositive

If $3n+2$ is odd, then n is also odd.

Analysis	**Proof by Contrapositive of If-then Theorem**
	• Restate the statement in its equivalent contrapositive form. • Use direct proof on the contrapositive form. •<u>State the origin statement as the conclusion.</u>

Contraposition and Compound Hypotheses

- Often a theorem has the form

$$(p \wedge q) \rightarrow r.$$

- This is logically equivalent to

$$\sim r \rightarrow \sim(p \wedge q).$$

- However, it is also equivalent to

$$p \rightarrow (q \rightarrow r).$$

  so it is also equivalent to

$$p \rightarrow (\sim r \rightarrow \sim q).$$

Form of Proof by Contradiction

- Theorem: $p \rightarrow q$.
- Outline of the proof of the theorem :
  - Assume $\sim(p \rightarrow q)$.
  - This is equivalent to assuming $p \wedge \sim q$.
  - Derive a contradiction, i.e., conclude $r \wedge \sim r$ for some statement r.
  - Conclude that $p \rightarrow q$.

Benefit of Proof by Contradiction

- The statement r may be *any statement whatsoever* because any contradiction

$$r \wedge \sim r$$

will suffice.

Proof By Contradiction Skeleton

- Suppose, for the sake of contradiction $\neg p$
- …
- q
- …
- $\neg q$
- But q and $\neg q$ is a contradiction! So we must have p.

Proof By Contradiction

- Claim: $\sqrt{2}$ is irrational (i.e. not rational).
- Proof:

Proof By Contradiction

- Claim: $\sqrt{2}$ is irrational (i.e. not rational).
- Proof:
- Suppose for the sake of contradiction that $\sqrt{2}$ is rational.

We don't have a fixed target.
That can make this proof harder.

- But [] is a contradiction!

Proof By Contradiction

If a^2 is even then a is e

- Claim: $\sqrt{2}$ is irrational (i.e. not rational).
- Proof:
- Suppose for the sake of contradiction that $\sqrt{2}$ is rational.
- By definition of rational, there are integers s, t such that $\mathrm{t} \neq 0$ and $\sqrt{2} = s/t$
- Let $p = \frac{\mathrm{s}}{\mathrm{gcd(s,t)}}, \mathrm{q} = \frac{t}{\mathrm{gcd(s,t)}}$ By the fundamental theorem of arithmetic, we have divided out a common factors of s, t and so p, q have no more common prime factors. Therefore the $\gcd(p, q$
- $\sqrt{2} = \frac{p}{q}$
- That's a contradiction! We conclude $\sqrt{2}$ is irrational.

Proof By Contradiction

If a^2 is even then a is e

- Claim: $\sqrt{2}$ is irrational (i.e. not rational).
- Proof:
- Suppose for the sake of contradiction that $\sqrt{2}$ is rational.
- By definition of rational, there are integers s, t such that $t \neq 0$ and $\sqrt{2} = s/t$
- Let $p = \frac{s}{\gcd(s,t)}, q = \frac{t}{\gcd(s,t)}$ By the fundamental theorem of arithmetic, we have divided out all factors of s, t and so p, q have no more common prime factors. Therefore the $\gcd(p, q) = 1$.
- $\sqrt{2} = \frac{p}{q}$
- $2 = \frac{p^2}{q^2}$
- $2q^2 = p^2$ so p^2 is even.
- That's a contradiction! We conclude $\sqrt{2}$ is irrational.

Proof By Contradiction

If a^2 is even then a is e

- Claim: $\sqrt{2}$ is irrational (i.e. not rational).
- Proof:
- Suppose for the sake of contradiction that $\sqrt{2}$ is rational.
- By definition of rational, there are integers s, t such that $t \neq 0$ and $\sqrt{2} = s/t$
- Let $p = \frac{s}{\gcd(s,t)}$, $q = \frac{t}{\gcd(s,t)}$ By the fundamental theorem of arithmetic, we have divided out all factors of s, t and so p, q have no more common prime factors. Therefore the $\gcd(p, q) = 1$.
- $\sqrt{2} = \frac{p}{q}$
- $2 = \frac{p^2}{q^2}$
- $2q^2 = p^2$ so p^2 is even. By the fact above, p is even, i.e. $p = 2k$ for some integer k. Squaring b $p^2 = 4k^2$
- Substituting into our original equation, we have: $2q^2 = 4k^2$, i.e. $q^2 = 2k^2$.
- So q^2 is even. Applying the fact above again, q is even.
- But if both p and q are even, $\gcd(p, q) \geq 2$. But we said $\gcd(p, q) = 1$
- That's a contradiction! We conclude $\sqrt{2}$ is irrational.

Proof By Contradiction

- How in the world did we know how to do that?

- In real life…lots of attempts that didn't work.
- Be very careful with proof by contradiction – without a clear target, you can easily end up in a loop of trying random things and getting nowhere.

Another Proof By Contradiction

- Claim: There are infinitely many primes.
- Proof:

Another Proof By Contradiction

- Claim: There are infinitely many primes.
- Proof:
- Suppose for the sake of contradiction, that there are only finitely many primes. Call them $p_1, p_2, \ldots, p_k$.

- But [] is a contradiction! So there must be infinitely many primes.

Another Proof By Contradiction

- Claim: There are infinitely many primes.
- Proof:
- Suppose for the sake of contradiction, that there are only finitely many primes. Call them $p_1, p_2, \ldots, p_k$.
- Consider the number $q = p_1 \cdot p_2 \cdot \cdots \cdot p_k + 1$
- Case 1: q is prime
- Case 2: q is composite
- But [] is a contradiction! So there must be infinitely many primes.

Another Proof By Contradiction

- Claim: There are infinitely many primes.
- Proof:
- Suppose for the sake of contradiction, that there are only finitely many primes. Call them $p_1, p_2, \ldots, p_k$.
- Consider the number $q = p_1 \cdot p_2 \cdot \cdots \cdot p_k + 1$
- Case 1: q is prime
- $q > p_i$ for all i. But every prime was supposed to be on the list $p_1, \ldots, p_k$. A contradiction!
- Case 2: q is composite
- Some prime on the list (say p_i) divides q. So $q \% p_i = 0$. and $(p_1 p_2 \cdots p_k + 1) \% p_i = 1$. But $q = (p_1 p_2 \cdots p_k + 1)$. That's a contradiction!
- In either case we have a contradiction! So there must be infinitely many primes.

Just the Skeleton

- “For all integers x, if x^2 is even, then x is even.”

Just the Skeleton

- “For all integers x, if x^2 is even, then x is even.”

- Suppose for the sake of contradiction, there is an integer x, such that x^2 is even and x is odd.
- …

- [] is a contradiction, so for all integers x, if x^2 is even, then x is even.

Just the Skeleton

- “There is not an integer k such that for all integers n, $k \geq n$.

Just the Skeleton

- “There is not an integer k such that for all integers n, $k \geq n$.
- Suppose, for the sake of contradiction, that there is an integer k such that for all integers n, $k \geq n$.
- …
- [] is a contradiction! So there is not an integer k such that for all integers $n, k \geq n$.

Contradiction vs. Contraposition

- Would this be a proof by contradiction or proof by contraposition?
- Proof by contraposition is preferred.

Contradiction vs. Contraposition?

- Theorem: Prove that if x is irrational, then $-x$ is also irrational.
- Proof:
  - Suppose that x is irrational and that $-x$ is rational.
  - Let $-x = a/b$, where a and b are integers.
  - Then $x = -(a/b) = (-a)/b$, which is rational.
  - This is a contradiction.
  - Therefore, if x is irrational, then $-x$ is also irrational.

What's the difference?

- What's the difference between proof by contrapositive and proof by contradiction?

Show $p \rightarrow q$	Proof by contradiction	Proof by contrap
Starting Point	$\neg(p \rightarrow q) \equiv (p \wedge \neg q)$	$\neg q$
Target	Something false	$\neg p$

Show p	Proof by contradiction	Proof by contrap
Starting Point	$\neg p$	---
Target	Something false	---

Counterexamples

- To disprove

$$\forall x\ P(x)$$

we simply need to find one number x in the domain of discourse that makes $P(x)$ false.

- Such a value of x is called a counterexample

Example 3

$$\forall n \in N,\ 2^n + 1 \text{ is prime}$$

Analysis	The statement is false

Example 4

$\boxed{\text{If } 3n+2 \text{ is odd, then } n \text{ is also odd.}}$